
Medical Image Registration

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Summary. Over the last decade, image registration has emerged as one of key technologies in medical image computing with applications ranging from computer assisted diagnosis to computer aided therapy and surgery. This chapter introduces the theory of medical image registration, its implementation and application. In particular, the three key components of any image registration algorithm are described: transformation models, similarity measures, and optimization techniques. The evaluation and validation of image registration is crucial in clinical applications and the chapter discusses techniques for the validation of image registration algorithms. Finally, the chapter illustrates the use of image registration techniques for image segmentation, shape modeling and clinical applications.

5.1 Introduction

Medical image registration [1] plays an increasingly important role in many clinical applications including Computer-assisted Diagnosis (CAD), Computer-aided Therapy (CAT) and Computer-assisted Surgery (CAS). A particular reason for the importance of image registration is the growing and diverse range of medical imaging modalities: Some modalities provide anatomical information about the underlying tissues such as the X-ray attenuation coefficient from X-ray Computed Tomography (CT) and proton density or proton relaxation times from Magnetic Resonance Imaging (MRI). Such images allow clinicians to quantify and visualize the size, shape and spatial relationship between anatomical structures and any pathology, if present. Other imaging modalities provide functional information such as the blood flow or glucose metabolism from Positron Emission Tomography (PET) or Single Photon Emission Computed Tomography (SPECT), and permit clinicians to study the relationship between anatomy and physiology. Finally, optical images acquired either in-vivo or ex-vivo (in form of histology) provide another important source of information which depicts structures at microscopic levels of resolution.

The goal of image registration is to find corresponding anatomical or functional locations in two or more images. Image registration can be applied to images from the same subject acquired by different imaging modalities (multi-modal image registration) or at different time points (serial image registration). Both cases are examples of intra-subject registration since the images are acquired from the same subject. Another application area for image registration is inter-subject registration where the aim is to align images acquired from different subjects, e.g., to study the anatomical variability within or across populations.

In general, the process of image registration involves finding the optimal geometric transformation which maximizes the correspondences across the images. This involves several components:

- A *transformation model* which defines a geometric transformation between the images. There are several classes of transforms including rigid, affine and deformable transforms. The question, which model is appropriate, is strongly linked to the application.
- A *similarity metric* or registration basis, which measures the degree of alignment between the images. In cases where features such as landmarks, edges or surfaces are available, the distances between corresponding features can be used to measure the alignment. In other cases, the image intensities can be used directly to determine the alignment.
- An *optimization method* which maximizes the similarity measure. Like many other problems in medical imaging, image registration can be formulated as an optimisation problem whose goal it is to maximize an associated objective function.
- A *validation protocol* which measures the performance of the registration techniques in general terms such as accuracy and robustness as well as in application-specific terms such as clinical utility.

In this chapter, we will discuss these components in more detail. We will also illustrate how image registration can be used in clinical applications. An extensive survey of registration techniques can be found in Zitova and Flusser [2]. Throughout this chapter, we will use $\mathbf{X}$, $\mathbf{x}$, and x to indicate a matrix, vector, or scalar quantity, respectively.

5.2 Transformation Model

The goal of image registration is to find a transformation $\mathbf{T} : (x, y, z) \mapsto (x', y', z')$, which maps any point in the source (or floating) image into the corresponding point in the target (or reference) image. There are several transformation models, ranging from simple (e.g., rigid or affine transformations) to more complex transforms (e.g., deformable transforms). The spectrum of simple to complex transforms is usually characterized by an increasing number of parameters that describe the Degrees of Freedom (DoF) of the transformation.

There are several key considerations which must be taken into account when choosing a transformation model:

1. It must be able to characterize the geometric transformation between the images. For example, if there is a significant amount of non-rigid deformation between the images, e.g., respiratory motion in images of the liver, then a transformation model only allowing for rigid or affine transforms is inappropriate as it cannot represent the transforms required to align the images.
2. It should be as simple as possible. For example, if two images are related by a simple rotation, then a non-rigid transformation is inappropriate as it has many parameters, which are not necessary to describe the solution.

5.2.1 Rigid Transformation

A common assumption in medical image registration is that both images are related by a rigid transformation. For example, for images of the head the rigid-body assumption is normally justified as the skull is rigid and constrains the motion of the brain sufficiently. In three dimensions, a rigid transformation involves six DoFs: three rotations and three translations. Using homogeneous coordinates [3], the rigid transformation can be expressed in matrix form

$$\mathbf{T}_{\text{rigid}}(x, y, z) = \begin{pmatrix} x' \\ y' \\ z' \\ 1 \end{pmatrix} = \begin{pmatrix} r_{11} & r_{12} & r_{13} & t_x \\ r_{21} & r_{22} & r_{23} & t_y \\ r_{31} & r_{32} & r_{33} & t_z \\ 0 & 0 & 0 & 1 \end{pmatrix} \begin{pmatrix} x \\ y \\ z \\ 1 \end{pmatrix} \quad (5.1)$$

where t_x, t_y, t_z define the translations along the axes of the coordinate system, while the coefficients r_{ij} are the result of the multiplication of three separate rotation matrices, which determine the rotations about each coordinate axis.

5.2.2 Affine Transformation

In some cases, it is necessary to correct not only for rigid transformation but also for scaling. This additional scaling can be expressed in matrix form as

$$\mathbf{T}_{\text{scale}} = \begin{pmatrix} s_x & 0 & 0 & 0 \\ 0 & s_y & 0 & 0 \\ 0 & 0 & s_z & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix} \quad (5.2)$$

where s_x, s_y , and s_z define the scaling along the different coordinate axes. In some cases, it may also be necessary to correct for shears, for example caused by the gantry tilt of CT scanners. A shear in the (x, y) plane can be expressed as

$$\mathbf{T}_{\text{shear}}^{xy} = \begin{pmatrix} 1 & 0 & h_x & 0 \\ 0 & 1 & h_y & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix} \quad (5.3)$$

Combining the rigid transformation matrix with the scaling and shearing matrices yields an affine transformation

$$\mathbf{T}_{\text{affine}}(x, y, z) = \mathbf{T}_{\text{shear}} \cdot \mathbf{T}_{\text{scale}} \cdot \mathbf{T}_{\text{rigid}} \cdot (x, y, z, 1)^T \quad (5.4)$$

whose twelve DoF represent rotations, translations, scaling and shears. Like rigid or affine transforms, global transforms affect the entire image domain. Higher-order global transforms such as tri-linear (24 DoF) or quadratic (30 DoF) transforms can be modeled in a similar fashion [4].

5.2.3 Projective Transformation

Projective transforms play an important role in applications involving the alignment of 3D volumes like CT or MRI to 2D images such as radiography and photography. Different types of projections including parallel or perspective projections can be used depending on the application [3]. However in most cases, the transformation that relates the 3D and 2D images is a combination of a projective with a rigid transformation, which determines the pose of the 3D volume relative to the camera. Often it is possible to determine the perspective transformation parameters using either knowledge about the internal geometry of the camera, or by camera calibration techniques [5], in which case the problem is reduced to rigid registration.

5.2.4 Non-Rigid Transformation: Parametric Models

The transformation models discussed so far can be characterized by a small number of parameters (six for rigid and twelve for affine transforms). While such transformation models are frequently used for the registration of anatomical structures like the brain or bones, they are not applicable in the case where significant deformation is expected, e.g., in soft tissues like the liver or breast. In these cases, deformable or non-rigid transforms are required. These non-rigid transforms can be modeled either using parametric or non-parametric models. In the case of non-parametric transformation models, a dense displacement field is stored which describes the deformation at every voxel. In contrast to this, parametric transformation models are controlled by a set of parameters (typically, the number of parameters is much smaller than the number of voxels).

Basis Functions

A common approach for modeling non-rigid transforms is to describe the transformation as a linear combination of basis functions θ_i :

$$\mathbf{T}(x, y, z) = \begin{pmatrix} x' \\ y' \\ z' \\ 1 \end{pmatrix} = \begin{pmatrix} a_{00} & \dots & a_{0n} \\ a_{10} & \dots & a_{1n} \\ a_{20} & \dots & a_{2n} \\ 0 & \dots & 1 \end{pmatrix} \begin{pmatrix} \theta_1(x, y, z) \\ \vdots \\ \theta_n(x, y, z) \\ 1 \end{pmatrix} \quad (5.5)$$

A common choice is to represent the deformation field using a set of (orthonormal) basis functions such as Fourier (trigonometric) [6, 7] or wavelet basis functions [8]. In the case of trigonometric basis functions this corresponds to a spectral representation of the deformation field where each basis function describes a particular frequency of the deformation. Restricting the summation in (5.5) to the first N terms (where $1 < N < n$) has the effect of limiting the frequency spectrum of the transformation to the N lowest frequencies.

Splines

Many registration techniques using splines are based on the assumption that a set of corresponding points or landmarks can be identified in the source and target images (see also the topic point-based registration in Sect. 5.3.1). These corresponding points are often referred to as control points. At these control points, spline-based transforms either interpolate or approximate the displacements, which are necessary to map the location of the control point in the target image into its corresponding counterpart in the source image. Between control points, they provide a smoothly varying displacement field. For each control point, the interpolation condition can be written as

$$\mathbf{T}(\phi_i) = \phi'_i \quad i = 1, \dots, n \quad (5.6)$$

where ϕ_i and ϕ'_i denote the location of the control point in the target and the source image, respectively. There are a number of different ways of how the control points can be determined. For example, anatomical or geometrical landmarks which can be identified in both images can be used [9]. In addition, Meyer et al. [10] suggested to update the location of control points by optimization of a voxel similarity measure such as mutual information. Alternatively, control points can be arranged with equidistant spacing across the image forming a regular mesh [11]. In this case, the control points are only used as a parameterization of the transformation and do not correspond to anatomical or geometrical landmarks. Therefore, they are often referred to as pseudo- or quasi-landmarks.

Thin-Plate Splines

Thin-plate splines are part of a family of splines which are based on radial basis functions. They have been formulated by Duchon [12] and Meinguet [13] for the surface interpolation of scattered data. In recent years, they have been

widely used for image registration [9, 14, 15]. Radial basis function splines can be defined as a linear combination of n radial basis functions $\theta(s)$:

$$t(x, y, z) = a_1 + a_2x + a_3y + a_4z + \sum_{j=1}^n b_j \theta(|\phi_j - (x, y, z)|) \quad (5.7)$$

Defining the transformation as three separate thin-plate spline functions $\mathbf{T} = (t_1, t_2, t_3)^T$ yields a mapping between images in which the coefficients $\mathbf{a}$ and $\mathbf{b}$ characterize the affine and the non-affine part of the transformation, respectively. The interpolation conditions in (5.6) form a set of $3n$ linear equations. To determine the $3(n+4)$ coefficients uniquely, twelve additional equations are required. These twelve equations guarantee that the non-affine coefficients $\mathbf{b}$ sum to zero and that their crossproducts with the x , y , and z coordinates of the control points are likewise zero. In matrix form, this can be expressed as

$$\begin{pmatrix} \Theta & \Phi \\ \Phi^T & \mathbf{0} \end{pmatrix} \begin{pmatrix} \mathbf{b} \\ \mathbf{a} \end{pmatrix} = \begin{pmatrix} \Phi' \\ \mathbf{0} \end{pmatrix} \quad (5.8)$$

Here $\mathbf{a}$ is a 4×3 vector of the affine coefficients $\mathbf{a}$, $\mathbf{b}$ is a $n \times 3$ vector of the non-affine coefficients $\mathbf{b}$, and Θ is the kernel matrix with $\Theta_{ij} = \theta(|\phi_i - \phi_j|)$. Solving for $\mathbf{a}$ and $\mathbf{b}$ using standard algebra yields a thin-plate spline transformation, which will interpolate the displacements at the control points.

The radial basis function of thin-plate splines is defined as

$$\theta(s) = \begin{cases} |s|^2 \log(|s|) & \text{in 2D} \\ |s| & \text{in 3D} \end{cases} \quad (5.9)$$

There are a wide number of alternative choice for radial basis functions including multi-quadrics and Gaussians [11, 16]. Modeling deformations using thin-plate splines has a number of advantages. For example, they can be used to incorporate additional constraints such as rigid bodies [17] or directional constraints [18] into the transformation model. They can be extended to approximating splines where the degree of approximation at the landmark depends on the confidence of the landmark localization [19].

Free-Form Deformations and B-Splines

In the late 1980s a number of techniques for modeling deformations emerged in the computer graphics community. In particular, Sederberg and Parry developed Free-Form Deformations (FFD) [20] as a powerful modeling tool for 3D deformable objects. The basic idea of FFDs is to deform an object by manipulating an underlying mesh of control points. The resulting deformation controls the shape of the 3D object and produces a smooth and continuous transformation. In the original paper by Sederberg and Parry [20], tri-variate Bernstein polynomials were used to interpolate the deformation between control points. A more popular choice is to use tri-variate B-spline tensor products

as the deformation function [21, 22]. The use of FFDs based on B-splines for image registration was first proposed by Rueckert et al. [23, 24]. Over the last decade, the use of FFDs for image registration has attracted significant interest [25–28].

To define a spline-based FFD, we denote the domain of the image volume as $\Omega = \{\mathbf{p} = (x, y, z) \mid 0 \leq x < X, 0 \leq y < Y, 0 \leq z < Z\}$. Let Φ denote a $n_x \times n_y \times n_z$ mesh of control points $\phi_{i,j,k}$ with uniform control point spacing δ (Fig. 5.1). Then, the FFD can be written as the 3D tensor product of the familiar 1D cubic B-splines:

$$T_{\text{local}}(\mathbf{p}) = \sum_{l=0}^3 \sum_{m=0}^3 \sum_{n=0}^3 B_l(u) B_m(v) B_n(w) \phi_{i+l, j+m, k+n} \quad (5.10)$$

where $i = \lfloor \frac{x}{\delta} \rfloor - 1, j = \lfloor \frac{y}{\delta} \rfloor - 1, k = \lfloor \frac{z}{\delta} \rfloor - 1, u = \frac{x}{\delta} - \lfloor \frac{x}{\delta} \rfloor, v = \frac{y}{\delta} - \lfloor \frac{y}{\delta} \rfloor, w = \frac{z}{\delta} - \lfloor \frac{z}{\delta} \rfloor$ and where B_l represents the l -th basis function of the B-spline [21, 22]:

$$\begin{aligned} B_0(u) &= (1 - u)^3/6 \\ B_1(u) &= (3u^3 - 6u^2 + 4)/6 \\ B_2(u) &= (-3u^3 + 3u^2 + 3u + 1)/6 \\ B_3(u) &= u^3/6 \end{aligned}$$

In contrast to thin-plate splines [15] or elastic-body splines [11], B-splines are locally controlled, which makes them computationally efficient even for a large number of control points. In particular, the basis functions of cubic B-splines have a limited support, i.e., changing control point $\phi_{i,j,k}$ affects the transformation only in the local neighborhood of that point.

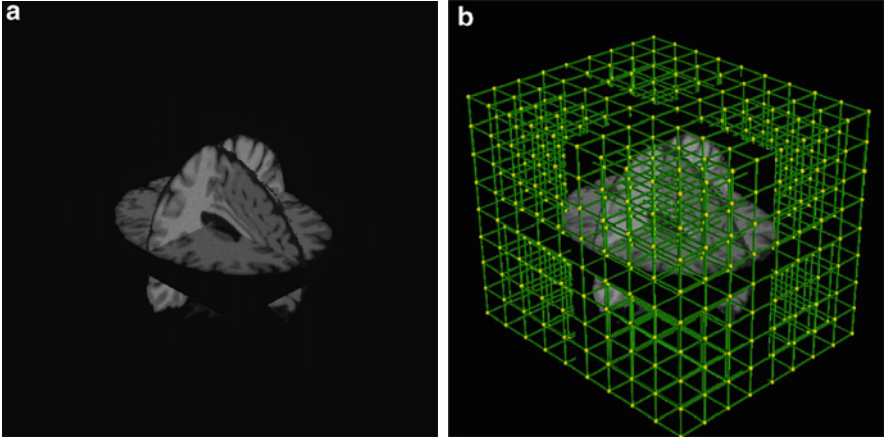


Fig. 5.1. *B-spline free-form deformations.* The axial, coronal and sagittal view of a 3D MRI of the brain is shown without control point mesh (a) and with control point mesh (b)

5.2.5 Non-Rigid Transformation: Non-Parametric Models

In contrast to parametric transformation models, non-parametric models characterize the deformation at every voxel. These models offer the greatest amount of flexibility in describing the transformation but are also expensive in terms of memory storage.

Small Deformation Models

Small deformation models encode the transformation at every voxel directly as a displacement vector. In computer vision, these models are closely related to the notion of optical flow [29], while in medical image registration their first use has been proposed in the context of elastic models [30]: The idea is to model the deformation of the source image into the target image as a physical process, which resembles the stretching of an elastic material such as rubber. This physical process is governed by two forces: The first term is the internal force, which is caused by the deformation of elastic material (i.e., stress) and counteracts any force, which deforms the elastic body from its equilibrium shape. The second term corresponds to the external force which acts on the elastic body. As a consequence, the deformation of the elastic body stops if both forces acting on the elastic body form an equilibrium solution. The behavior of the elastic body is described by the Navier linear elastic Partial Differential Equation (PDE):

$$\mu \nabla^2 \mathbf{u}(x, y, z) + (\lambda + \mu) \nabla(\nabla \cdot \mathbf{u}(x, y, z)) + \mathbf{f}(x, y, z) = 0 \quad (5.11)$$

Here $\mathbf{u}$ describes the displacement field, $\mathbf{f}$ is the external force acting on the elastic body; ∇ and ∇^2 denote the gradient operator and the Laplace operator, respectively. The parameters μ and λ are Lamé's elasticity constants, which describe the behavior of the elastic body. These constants are often interpreted in terms of Young's modulus E_1 , which relates the strain and stress of an object, and Poisson's ratio E_2 , which is the ratio between lateral shrinking and longitudinal stretching:

$$E_1 = \frac{\mu(3\lambda + 2\mu)}{(\lambda + \mu)}, \quad E_2 = \frac{\lambda}{2(\mu + \lambda)} \quad (5.12)$$

The external force $\mathbf{f}$ is the force which acts on the elastic body and drives the registration process. A common choice for the external force is the gradient of a similarity measure such as a local correlation measure based on intensities [30], intensity differences [31] or intensity features such as edge and curvature [32], however, in principle any of the similarity metrics described in Sect. 5.3 can be used as a force.

The PDE in (5.11) may be solved by finite differences and Successive Over-Relaxation (SOR) [33]. This yields a discrete displacement field for each voxel. Alternatively, the PDE can be solved for only a subset of voxels which

correspond to the nodes of a finite element model [32, 34]. These nodes form a set of points for which the external forces are known. The displacements at other voxels are obtained by finite element interpolation. An extension of the elastic registration framework has been proposed by Davatzikos [35] to allow for spatially varying elasticity parameters. This enables certain anatomical structures to deform more freely than others.

Large Deformation Model

In the small deformation model, the displacements are stored at each voxel with respect to their initial position. Since the regularization in (5.11) acts on the displacement field, highly localized deformations cannot be modeled since the deformation energy caused by stress increases proportionally with the strength of the deformation. In the large deformation model, however, the displacement is generated via a time dependent velocity field $\mathbf{v}$ [36]. The relationship between $\mathbf{v}$ and the displacement $\mathbf{u}(x, y, z, 1)$ is given by

$$\mathbf{u}(x, y, z, 1) = \int_0^1 \mathbf{v}(\mathbf{u}(x, y, z, t)) dt \quad (5.13)$$

with $\mathbf{u}(x, y, z, 0) = (x, y, z)$. The resulting transformations are *diffeomorphic* as they provide a smooth one-to-one (invertible) mapping. One of the first approaches to use such a representation of the deformations coined the term fluid registration [37]. Here the deformations are characterized by the Navier–Stokes partial differential equation

$$\mu \nabla^2 \mathbf{v}(x, y, z) + (\lambda + \mu) \nabla(\nabla \cdot \mathbf{v}(x, y, z)) + \mathbf{f}(x, y, z) = 0 \quad (5.14)$$

similar to (5.11) except that differentiation is carried out on the velocity field $\mathbf{v}$ rather than on the displacement field $\mathbf{u}$ and is solved for each time step. Christensen [37] suggested to solve (5.14) using SOR [33]. However, the resulting algorithm is rather slow and requires significant computing time. A faster implementation has been proposed by Bro-Nielsen et al. [38]. Here, (5.14) is solved by deriving a convolution filter from the eigenfunctions of the linear elasticity operator. Bro-Nielsen et al. [38] also pointed out that this is similar to a regularization by convolution with a Gaussian as proposed in a non-rigid matching technique by Thirion [39], in which the deformation process is modeled as a diffusion process. However, the solution of (5.14) by convolution is only possible if the viscosity is assumed constant, which is not always the case.

5.3 Registration Basis

The second component of a registration algorithm is the registration basis, which measures the degree of alignment of the images. The two main approaches are feature-based and voxel-based similarity measures.

5.3.1 Feature-Based Registration

Feature-based registration approaches are usually based on points, lines or surfaces and aim at minimizing the distance between corresponding features in the two images. This requires the extraction of the features as well as the estimation of correspondences.

Points

One of the most intuitive registration criteria is the proximity of corresponding point features, which can be identified in the two images. Given two point sets $\mathbf{p}$ and $\mathbf{q}$, we can define a similarity measure based on the squared distance of the points:

$$\mathcal{S} = - \sum_i \|\mathbf{q}_i - \mathbf{f}(\mathbf{p}_i)\|^2 \quad (5.15)$$

In (5.15), we assume that the correspondences across the point sets are known a priori (we will see later how to formulate this problem if correspondences are unknown, e.g., aligning surfaces described by point clouds). For a rigid transformation, three or more non collinear landmarks are sufficient to establish the transformation between two 3D images. The optimization problem defined in (5.15) can be solved in closed-form [40] and does not require any iterative optimization techniques: It involves the alignment of centroids of the two sets of points followed by a rotation to minimize the sum of the squared displacements between source and destination points. This is achieved by simple matrix manipulation using the method of Singular Value Decomposition (SVD) [33].

Points that can be used for registration can be intrinsic, anatomical landmarks which correspond to point-like structures (e.g., apical turn of the cochlea) or to points can be unambiguously defined (e.g., junction of the vertebral arteries or center of the orbit of the eyes). Other possibilities include the identification of geometrical features of anatomical structures such as a maximum of 2D curvature (e.g., cartoid syphon) or a maximum of 3D curvature (e.g., occipital pole of the brain). A disadvantage in using intrinsic landmarks is the sparse nature and the difficulty of automatic identification and extraction of these landmarks, even though several approaches exist [19]. Alternatively, one can use extrinsic landmarks such as pins or markers fixed to the patient and visible on each image. These markers may be attached to the skin or screwed into the bone. The latter can provide very accurate registration but are more invasive and uncomfortable for the patient. Skin markers on the other hand can easily move by several millimeters due to the mobility of the skin and are difficult to attach firmly. However, well-designed, extrinsic markers can be accurately localized with sub-voxel precision, for instance, by computing the center of gravity of spherical or cylindrical markers [41]. The task of identifying the markers can be automated using computer vision techniques [42].

Surfaces

Point features are relatively sparse and for many tasks, in particular for non-rigid registration, more dense features are required. Another feature that can be used for the registration of images are the boundaries of corresponding anatomical structures. This requires the segmentation of a contour or surface representing the boundary of the anatomical structure in both images and can be achieved using an interactive or automated procedure. The resulting contours or surfaces are often represented as point sets, which can be registered by minimizing the distance between corresponding points of both sets [43, 44]. However, since the correspondences between both point sets are not known a priori, these need to be estimated at the same time. Besl and McKay proposed a generic registration algorithm for point sets, called Iterative Closest Point (ICP) algorithm [45], which assumes that there is a correspondence between each point in the first set and its closest point in the second set. In this approach, the similarity measure can be defined as

$$\mathcal{S} = - \sum_i ||\mathbf{y}_i - \mathbf{T}(\mathbf{p}_i)||^2 \quad (5.16)$$

where

$$\mathbf{y}_i = \min_{\mathbf{q}_j \in \mathbf{q}} \{ ||\mathbf{q}_j - \mathbf{T}(\mathbf{p}_i)||^2 \} \quad (5.17)$$

is the closest point in $\mathbf{q}$. To register two surfaces in form of point sets we first estimate correspondences between the point sets using (5.17). Then, the resulting point sets can be registered by minimizing (5.16), e.g., using the point-based technique discussed in the previous section. The process is repeated until convergence is achieved. In practice, the computational cost can be significantly reduced by using efficient spatial data structures such as octrees [46] or distance transforms [47, 48] to pre-calculate the distance to the closest point.

5.3.2 Voxel-Based Registration

The advantage of feature-based registration is that it can be used for both mono- and multi-modality registration. However, the disadvantage of feature-based registration is the need for feature extraction in form of landmark detection or segmentation. Moreover, while features can be extracted manually or automatically, any error during the feature extraction stage will propagate into the registration and cannot be recovered at a later stage. To avoid these errors, it is possible to use the image intensities directly without the need for feature extraction. In this case, voxel-based similarity measures aim at measuring the degree of shared information of the image intensities. This is relatively simple in the case of mono-modality registration, but more complex

for multi-modality registration. Over the last decade, voxel-based similarity measures have become the method of choice for measuring image alignment, largely due to their robustness and accuracy.

Statistical Similarity Measures

The simplest statistical measure of image similarity is based on the Sum of Squared Differences (SSD) between the intensities in the images $\mathcal{I}_A$ and $\mathcal{I}_B$,

$$\mathcal{S}_{\text{SSD}} = -\frac{1}{n} \sum (\mathcal{I}_A(\mathbf{q}) - \mathcal{I}_B(\mathbf{T}(\mathbf{p})))^2 \quad (5.18)$$

where n is the number of voxels in the region of overlap. This measure is based on the assumption that both imaging modalities have the same characteristics. If the images are correctly aligned, the difference between them should be zero except for the noise produced by the two modalities. If this noise is Gaussian distributed, it can be shown that the SSD is the optimal similarity measure [49]. Since this similarity measure assumes that the imaging modalities are identical, it is restricted to mono-modal applications such as serial registration [50, 51].

In a number of cases, the assumption of identical imaging modalities is too restrictive. A more general assumption is that of a linear relationship between the two images. In these cases, the similarity between both images can be expressed by the Normalized Cross Correlation (NCC)

$$\mathcal{S}_{\text{NCC}} = \frac{\sum (\mathcal{I}_A(\mathbf{q}) - \mu_A)(\mathcal{I}_B(\mathbf{T}(\mathbf{p})) - \mu_B)}{\sqrt{(\sum \mathcal{I}_A(\mathbf{q}) - \mu_A)^2 (\sum \mathcal{I}_B(\mathbf{T}(\mathbf{p})) - \mu_B)^2}} \quad (5.19)$$

where μ_A and μ_B correspond to average voxel intensities in the images $\mathcal{I}_A$ and $\mathcal{I}_B$, respectively. Nevertheless, the application of this similarity measure is largely restricted to mono-modal registration tasks.

Information-Theoretical Measures

There has been significant interest in measures of alignment based on the information content or entropy of the registered image. An important step to understanding these methods is the feature space of the image intensities which can also be interpreted as the joint probability distribution: A simple way of visualizing this feature space is by accumulating a 2D histogram of the co-occurrences of intensities in the two images for each trial alignment. If this feature space is visualized for difference degrees of image alignment, it can be seen that the feature space disperses as misalignment increases, and that each image pair has a distinctive feature space signature at alignment.

In an information theoretic framework the information content of images can be defined as the Shannon–Wiener entropy, $H(\mathcal{I}_A)$ and $H(\mathcal{I}_B)$ of images

$\mathcal{I}_A$ and $\mathcal{I}_B$:

$$H(\mathcal{I}_A) = - \sum_{a \in \mathcal{I}_A} p(a) \log p(a) \quad (5.20)$$

and

$$H(\mathcal{I}_B) = - \sum_{b \in \mathcal{I}_B} p(b) \log p(b) \quad (5.21)$$

where $p(a)$ is the probability that a voxel in image $\mathcal{I}_A$ has intensity a and $p(b)$ is the probability that a voxel in image $\mathcal{I}_B$ has intensity b . The joint entropy $H(\mathcal{I}_A, \mathcal{I}_B)$ of the overlapping region of image $\mathcal{I}_A$ and $\mathcal{I}_B$ may be defined by

$$H(\mathcal{I}_A, \mathcal{I}_B) = - \sum_{a \in \mathcal{I}_A} \sum_{b \in \mathcal{I}_B} p(a, b) \log p(a, b) \quad (5.22)$$

where $p(a, b)$ is the joint probability that a voxel in the overlapping region of image $\mathcal{I}_A$ and $\mathcal{I}_B$ has values a and b , respectively.

To derive a measure of image alignment one can use concepts from information theory such as Mutual Information (MI) [52, 53]. In terms of entropies, MI is defined as

$$\mathcal{S}_{MI}(\mathcal{I}_A; \mathcal{I}_B) = H(\mathcal{I}_A) + H(\mathcal{I}_B) - H(\mathcal{I}_A, \mathcal{I}_B) \quad (5.23)$$

and should be maximal at alignment. Mutual information is a measure of how one image “explains” the other but makes no assumption of the functional form or relationship between image intensities in the two images. It is also not independent of the overlap between two images [54]. To avoid any dependency on the amount of image overlap, Normalised Mutual Information (NMI) has been suggested as a measure of image alignment [54]:

$$\mathcal{S}_{NMI}(\mathcal{I}_A; \mathcal{I}_B) = \frac{H(\mathcal{I}_A) + H(\mathcal{I}_B)}{H(\mathcal{I}_A, \mathcal{I}_B)} \quad (5.24)$$

Similar forms of normalised mutual information have been proposed by Maes et al. [55].

Information-theoretic voxel similarity measures are based on the notion of the marginal and joint probability distributions of the two images. These probability distributions can be estimated in two different ways: The first method uses histograms whose bins count the frequency of occurrence (or co-occurrence) of intensities. Dividing these frequencies by the total number of voxels yields an estimate of the probability of that intensity. The second method is based on generating estimates of the probability distribution using Parzen windows [56], which is a non-parametric technique to estimate probability densities. The Parzen-based approach has the advantage of providing a differentiable estimate of mutual information, which is not the case for the histogram-based estimate of mutual information.

5.4 Optimization

Like many other problems in medical image analysis, the problem of image registration can be formulated as an optimization problem whose goal is to maximize an objective function. In general, the objective function can be written as combination of two terms: The first term aims at maximizing the similarity of the images and is the driving force behind the registration process. The second term aims at minimizing the cost associated with particular transformations. In the case of rigid or affine registration, the second term usually plays no role and is often omitted. In this case, maximizing the objective function is equal to maximizing the similarity metric. However, in the case of non-rigid registration, the second term can act as a regularization or penalty function which can be used to constrain the transformation relating both images: For example, in elastic or fluid registration the regularization term (linear-elasticity model) forms an integral part of the registration. Other regularization models are the Laplacian or membrane model [6, 57] and the biharmonic or thin-plate model [15, 58]. Both models have an intuitive physical interpretation: While the Laplacian model approximates the energy of a membrane (such as a rubber sheet), which is subjected to elastic deformations, the biharmonic term approximates the energy of a thin plate of metal which is subjected to bending deformations [59]. From a probabilistic point of view, the regularization term can be interpreted as a prior which represents a priori knowledge about the expected transformation, whereas the similarity metric can be viewed as a likelihood term that expresses the probability of a match between the images.

5.5 Validation of Registration

Prior to clinical use, medical image registration algorithms need to be validated. However, the validation of registration performance usually suffers from the lack of knowledge as to whether, how much, and where patient movement has occurred between and even during scanning procedures, and whether such movement affects the clinical usefulness of the data. To maintain clinical usefulness, and inherently improve patient treatment and health care, it is, therefore, mandatory to ensure that registration is successful [60].

A registration method can be assessed in an independent evaluation in the absence of a ground truth. An initial visual inspection allows for a qualitative assessment of registration performance, which can be complemented by quantitative checks for robustness and consistency. The former establishes the measurement precision by testing the bias sensitivity when adding noise or choosing different starting estimates [61]. The latter assesses the capability of a registration technique to find circular transforms based on a registration circuit, but can be sensitive to bias and may not be applicable to non-invertible transforms generated by many non-rigid registration methods. Nonetheless,

consistency checks have been successfully used for intra-modality rigid body registration applications (e.g., serial MRI of the brain [62]). In addition, an expert observer can perform a visual assessment of the registration performance. This can involve the inspection of subtraction images, contour or segmentation overlays, alternate pixel displays, or viewing anatomical landmarks. These approaches have been applied to rigid registration [63], and since they involve inspection of the entire volume domain of the image pair, they can be extended to non-rigid registration [64]. For non-rigid registration, visual assessment is an important step toward clinical acceptance and routine use of a registration method, but may be compromised by locally implausible deformations which may not be readily picked up by observers [65]. Nonetheless, visual assessment often forms the first and last line of defense of any image registration validation.

In the absence of ground truth, registration accuracy can be studied by setting up a gold standard transformation. For example, the Retrospective Registration Evaluation Project (RREP) used skull-implanted markers in patients undergoing brain surgery to derive a gold standard for multi-modal rigid-body image registration of the head to compare different established rigid registration methods [66]. For non-rigid registration validation, extrinsic markers cannot easily be used as they would need to be implanted into soft tissue. In a recent study by Koshani et al. [67], markers were inserted into a CT lung phantom, which was deformed and imaged under controlled conditions. In an alternative approach [68], a bio-mechanical motion simulator was introduced that allowed the simulation of physically plausible deformations for clinically realistic motion scenarios, with application to contrast-enhanced Magnetic Resonance Mammography (MRM). This motion simulator was designed to be independent of the image registration and transformation model used. Similar approaches have been proposed for the simulation of X-ray type compression for MRM [69] and Alzheimer's Disease [70].

Finally, the segmentation of anatomical structures provide the means to measure the structure overall or surface distances before and after registration, but does not provide any insight as to whether the registration is accurate within the structure, or along its outline. If, however, the objective of the registration is to propagate (and hence automate) segmentation, the segmentation quality can be used as a surrogate measurement. For example in [71], a number of non-rigid registration methods were compared for inter-subject brain registration including segmentation quality. A number of carefully annotated image databases¹ are emerging, which can be used to establish the accuracy of non-rigid registration methods on the basis of carefully delineated image structures.

¹ <http://www.nirep.org/>

5.6 Application

There are numerous applications of image registration ranging from clinical applications and trials to image registration as a tool for other image analysis tasks such as segmentation or shape modeling.

5.6.1 Intra-Subject Registration

Intra-subject registration applications can be divided into registering images of the same subject acquired using different imaging modalities (multi-modal registration or fusion) or at different time points (serial registration).

Multi-Modal Image Fusion

Medical imaging modalities can be broadly distinguished into two categories: anatomical and functional. In many clinical applications, imaging modalities from both categories must be combined for an effective diagnosis. One such example is the registration of PET with either CT or MRI for applications in oncology. Here, the PET is used to determine the tumor activity and the CT/MRI allows accurate localization of this activity. Similarly, in many neurological diseases the metabolism in the brain is altered by the disease. This can be detected using PET and localized using CT/MRI. Multi-modal registration is usually based on features such as points and surfaces [43] or uses information theoretic similarity measures such as mutual information [72, 73].

Quantification of Changes

A key application of image registration is the alignment of images from the same subject acquired at different time points. The difference between successive time points can range from fractions of a second (e.g., cardiac motion) to several years (e.g., longitudinal growth, atrophy). The comparison of images across time enables the quantification of the anatomical differences between the time points of measurement. These differences can have a number of reasons, such as bulk patient motion, organ motion (e.g., respiratory or cardiac motion), growth, atrophy, or disease progression.

In neuroimaging, image registration has been widely used for the detection of changes in the human brain: For example, the rigid registration of serial MRI is well suited for the visualization of subtle changes in the brain [50, 51]. Moreover, it is also possible to quantify the amount of changes by using non-rigid registration of serial MRI [74, 75]: When two images are related to each other by a non-rigid transformation, the target image is subjected to local deformation, which locally changes the volume of regions in the coordinate space. The local volume change in an infinitesimally small neighborhood around any given point is represented by the local Jacobian determinant of the coordinate transformation at that point

$$|J(x, y, z)| = \det \begin{pmatrix} \frac{\partial T_x}{\partial x} & \frac{\partial T_x}{\partial y} & \frac{\partial T_x}{\partial z} \\ \frac{\partial T_y}{\partial x} & \frac{\partial T_y}{\partial y} & \frac{\partial T_y}{\partial z} \\ \frac{\partial T_z}{\partial x} & \frac{\partial T_z}{\partial y} & \frac{\partial T_z}{\partial z} \end{pmatrix} \quad (5.25)$$

If there is no local volume change, $J = 1$. If there is a local volume decrease or increase, $J < 1$ or $J > 1$, respectively. Thus, local volume change can be estimated directly from the deformation field that aligns baseline and follow-up images. In such a way, growth patterns have been mapped in 4D (3D plus time) in older children [76, 77], and clinically useful information has been extracted in conditions such as Alzheimer's Disease [78, 79], and imaginative studies of neurogenetics and development have been undertaken [80].

5.6.2 Inter-Subject Registration

In cross-sectional studies, non-rigid registrations between images of different subjects can be used to characterize the differences between populations or the differences between an individual and a reference image. Approaches that focus on the transformations in this way have been referred to as Voxel-Based Morphometry (VBM), Deformation-Based Morphometry (DBM) or Tensor-Based Morphometry (TBM).

In a simple cross-sectional study, for example, where two clinical groups are to be separated, it is possible to non-rigidly register all the images to a reference image. If the non-rigid registration only accounts for global shape, differences methods such as voxel-based morphometry [81] can be used to investigate the differences in the probabilistic segmentations. If, however, the non-rigid registration also accounts for local shape differences, the differences between the subjects are encoded in the resulting transforms and their properties can be used to identify group differences. Such approaches usually use properties such as the Jacobian of the deformation fields as defined in eq. (5.25) to analyze these differences and is often referred to as DBM or TBM [82].

Segmentation

The amount of data produced by imaging increasingly exceeds the capacity for expert visual analysis, resulting in a growing need for automated image analysis. In particular, accurate and reliable methods for segmentation (classifying image regions) are a key requirement for the extraction of information from images, cf. Chap. 11, page 279. A common approach to automatic segmentation is atlas-based segmentation in which an anatomical atlas is registered to an individual's anatomical image [83, 84]. By transforming the segmentation of the atlas into the coordinate system of the subject, one can obtain a segmentation of the subject's image. Errors in the registration process will affect the accuracy of the propagated segmentation.

More recently, it has been shown that using multiple atlases and combining the multiple segmentations obtained by registration, random errors in the registration can be compensated for, resulting in an improved segmentation [85]. Using this method, each atlas is registered to the subject in question. The resulting transformation is then used to transform the atlas segmentation into the subject's coordinate system. By applying decision fusion techniques at every voxel in subject space, e.g., using the vote rule [86] or Simultaneous Truth And Performance Level Estimation (STAPLE) [87], an average segmentation can be generated (Fig. 5.2).

Shape Modeling

Statistical models of shape variability have been successfully applied to perform various analysis tasks in 2D and 3D images. In particular, their application for image segmentation in the context of Active Shape Models (ASMs) has been very successful [88]. In building those statistical models, a set of segmentations of the shape of interest is required as well as a set of landmarks that can be unambiguously defined in each sample shape. However, the manual identification of such correspondences is a time consuming and tedious task. This is particularly true in 3D where the amount of landmarks required to describe the shape accurately increases dramatically as compared

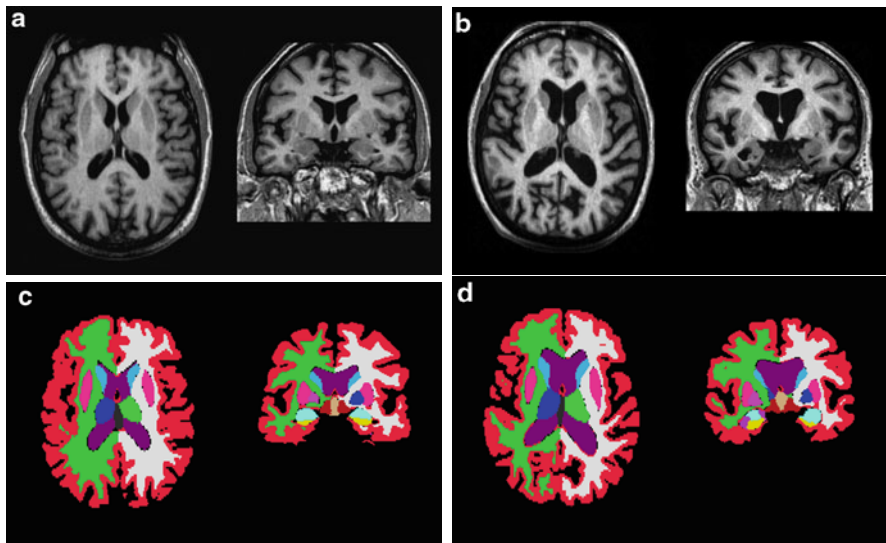


Fig. 5.2. *Example of multi-atlas segmentation in brain MRI [85]. (a)* Normal elderly subject and (c) corresponding segmentation. (b) subject with Alzheimer's Disease and (d) corresponding segmentation. Despite the significant differences between normal and pathologic anatomy, the segmentation produced by multi-atlas fusion is stable

to 2D applications. While several approaches have been proposed that aim at addressing this problem by automating the process of correspondence estimation [89–91], the problem of finding correspondences for the construction of shape models can be solved via registration. For example, Frangi et al. [92] have proposed an procedure in which an atlas or reference shape is registered to all other shapes and the landmarks of the atlas are propagated to all other shapes. The registration used for this purpose can be either surface-based [93] or image-based registration [92].

A complementary approach to building statistical shape models of anatomical variability is the concept of statistical deformation models, which is closely related to the rapidly developing discipline on computational anatomy pioneered by the work of Grenander and Miller [94]. One of the key ideas here is to carry out a statistical analysis directly on the deformation fields that describe a dense correspondence between the anatomies. There are two distinct advantages of this approach: First, the resulting statistical models are not limited to single anatomical structures but can instead describe the intra- and inter-structure variability of the anatomy across a population. Second, the deformation fields can be obtained by non-rigid registration, thereby eliminating the need for any segmentation. Such an approach has been developed by several groups [95–97].

5.7 Summary and Conclusions

In this chapter, we have presented a number of theoretical and practical aspects of medical image registration. Medical image registration is widely used, both in clinical applications (e.g., image fusion, image-guided surgery) as well as a tool for biomedical research (e.g., to study populations in clinical trials). The components of a registration method very much depend on the clinical application at hand: The majority of applications in clinical practice uses rigid or affine registration where automated solutions have been shown to be accurate, robust and fast. The transformation model in rigid/affine registration is obviously fixed, but current developments focus on finding better registration bases (similarity measures, in particular for multi-modal or contrast-enhanced imaging), as well as faster optimization techniques for automated real-time image guidance. There is an increasing interest in non-rigid registration for clinical applications. However, automated solutions have not yet reached the same degree of maturity as for rigid or affine registrations. non-rigid registration is still an area of on-going research and most algorithms are in the stage of development and evaluation.

A particular development focus is on diffeomorphic transforms, which allow a one-to-one (invertible) mapping between different subjects, and more accurate intra-subject registration in dynamic imaging, such as contrast-enhanced MRI. For physically more plausible transforms, local constraints are investigated in an attempt to preserve volume locally, and ensure smooth

deformation fields. Biomechanical motion models are limited by lack of a priori knowledge about patient-specific tissue properties, and are generally computationally very complex. The lack of a generic gold standard for assessing and evaluating the success of non-rigid registration algorithms is mostly performed using volume overlaps, sparse landmark distances or surface distances of delineated structures, but non-rigid validation methodologies have been emerging using biomechanical motion or disease simulations [68, 70] or physical phantoms [98] for specific clinical applications.

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